

Geometric Ergodicity of the Restricted Metropolis Algorithm

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Abstract.

For an ergodic Metropolis-Hastings algorithm satisfying some compactness condition, we show that the induced Markov chain has a spectral gap if and only if the probability of rejection is bounded away from unity. Using this result, we propose a modification of the Metropolis algorithm (that we called the Restricted Metropolis algorithm) which is shown to be uniformly geometrically ergodic for any density (bounded on compact sets) with tail lighter than the proposal distribution's tail. A simulation example is presented to illustrate. *AMS Classification:* 65C05, 65C40, 60J27, 60J35.

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1. Introduction

The Metropolis-Hastings (MH) algorithm initiated by ? is a very flexible algorithm used to approximately sample from complicated distributions in high dimension spaces.

If π is the probability distribution of interest, such an algorithm generates a Markov chain (X_n) which admits π as its stationary distribution. If the chain is ergodic, then for n sufficiently large, X_n is taken as approximately distributed as π . From a statistical point of view, geometric ergodicity is a desirable property for such algorithms. Specifically, under geometric ergodicity, the Central Limit Theorem for empirical sum of functions of X_n is (theoretically) easy to check as in ?. Geometric ergodicity can also be used to design convergence diagnostic techniques, a big issue in practical use of MCMC methods (see ?). There is an interesting discussion about the usefulness of geometric ergodicity of MCMC algorithms in ?.

In section 1, we show that under a compacity condition, an ergodic Metropolis-Hastings chain is geometrically ergodic if the rejection probability of the chain is bounded away from unity. As we shall see, this result can be useful to design new algorithms which are geometrically ergodic. Actually, it is a partial converse of ? who proved that if a Metropolis-Hastings chain is geometrically ergodic, then the rejection probability of the chain is bounded away from unity.

Our techniques of proof differ from those in ? as we mainly use results from Hilbert spaces operators theory. We decompose K_0 , the operator induced by the Metropolis-Hastings chain, as the sum of a multiplication operator M_r (multiplication by the rejection probability or the chain denoted r) and an integral operator U . Under a compacity condition on U , the Weyl's theorem (?) states that the “continuous” part of the spectrum of K_0 is the same with that of M_r which is $\text{ess-ran}(r)$, the essential range of r . Then the reversibility and the ergodicity of the chain is used to assure that the “discrete” part of the spectrum of K_0 remains bounded away from unity.

In the particular case of the Independent Metropolis-Hastings algorithm (IMH), the compacity condition is always satisfied, but in this case more can be said. In theorem (2.2), we show that the spectrum of the IMH chain is exactly the essential range of the rejection probability of the chain. ? proved this result when the state space of the chain is finite or discrete (with an additional regularity condition on π). He has also conjectured that the result holds in general spaces. ? have derived the transition kernel of the successive iterates of the IMH chain in general space. But their results did not clearly solve this conjecture.

In general, it may be difficult to choose a good proposal to use with the MH algorithm. This is certainly one reason why the Metropolis algorithm is still widely used. But as shown by ?, exponential or lighter tail is necessary for the Metropolis algorithm to be geometrically ergodic. For heavy tailed distribution, polynomial rate of convergence is possible. In this direction, ? have shown that for a symmetric target density with polynomial tail, and for any proposal density with tail that also recedes polynomially, the Metropolis algorithm has a polynomial rate of convergence.

In section 3, as another solution, we propose to modify the Metropolis algorithm by restricting the random walk behaviour on a compact set. This restriction yields a chain that is geometrically ergodic whenever the target density have a lighter tail than the proposal's tail. The idea is as follow. We fix a convex compact set Δ . Ideally, Δ contains all the modes of π . At any $x \in \Delta$, the proposal move is made from a distribution centered at x as in the Metropolis algorithm. But when the chain reaches $x \notin \Delta$, the proposal distribution is centered at $p(x)$ the orthogonal projection of x on Δ . As an application of theorem (2.1), we show in theorem (3.1) that this algorithm is geometrically

ergodic for any density (bounded on compact sets) with tail decaying faster than the proposal's tail. In theorem (3.2), using Markov chains techniques, we strengthen the conclusions of theorem (3.1) by showing that this Restricted Metropolis algorithm is uniformly geometrically ergodic if and only if the above mentioned tail condition hold.

The main points of the paper are as follow. The geometric ergodicity of the general Metropolis-Hastings algorithm is shown in theorem (2.1) of section 2. The spectrum of the IMH chain is derived in theorem (2.2). The Restricted Random Walk Metropolis algorithm is introduced in section 3. Its rate of convergence is investigated in theorems (3.1) and (3.2) of section 3. A simulation example is also presented to illustrate the benefit of restricting the Metropolis algorithm.

2. Geometric Ergodicity of the Metropolis-Hastings Chain

Throughout the section, we fix $(\mathcal{S}, \mathcal{F}, \pi)$ a probability space, where π is our target probability measure. Let $Q(x, \cdot)$ be a transition kernel on $\mathcal{S}$. We shall assume that for all x , $Q(x, \cdot)$ is absolutely continuous with respect to π and we write $Q(x, dy) = \omega(x, y)\pi(dy)$.

Before going further, we recall the Metropolis-Hastings algorithm for simulating a Markov chain which admits π as its invariant distribution.

Algorithm 2.1. The Metropolis-Hastings Algorithm:

At the i th iteration, $X_i = x$.

Generate $Y_{i+1} = y$ from $Q(x, \cdot)$.

set

$$X_{i+1} = \begin{cases} y & \text{with probability } \alpha(x, y) \\ x & \text{with probability } 1 - \alpha(x, y) \end{cases}$$

With

$$\alpha(x, y) = \begin{cases} \min\left(1, \frac{\omega(y, x)}{\omega(x, y)}\right) & \text{if } \omega(x, y) \neq 0 \\ 1 & \text{if } \omega(x, y) = 0 \end{cases}$$

The algorithm leaves a lot of room in choosing Q . When $Q(x, \cdot) = Q(\cdot)$ for all $x \in \mathcal{S}$, the algorithm thus obtained is usually called the Independent Metropolis-Hastings algorithm (IMH). If $\mathcal{S} = \mathbb{R}^d$ and $Q(x, \cdot)$ has a density with respect to the Lebesgue measure with $Q(x, dy) = q(|y - x|)dy$, we obtain the Random Walk Metropolis algorithm (RWMH), where $|x|$ denotes the norm of $x \in \mathcal{S}$.

The algorithm (2.1) generates a Markov chain (X_n) with transition kernel

$$P(x, dy) = \alpha(x, y)\omega(x, y)\pi(dy) + r(x)\delta_x(dy), \quad (2.1)$$

where

$$r(x) = 1 - \int \alpha(x, y)\omega(x, y)\pi(dy) \quad (2.2)$$

is the probability of rejection of the chain and $\delta_x(A) = 1$ if $x \in A$ and 0 otherwise. It is well known that P is reversible with respect to π , that is:

$$\pi(dx)P(x, dy) = \pi(dy)P(y, dx), \quad (2.3)$$

as measures on $\mathcal{S} \times \mathcal{S}$. This implies that P admits π as an invariant distribution, $\pi P = \pi$ where

$$\pi P(A) := \int \pi(dx) P(x, A). \quad (2.4)$$

Below, we first review some basic concepts from Markov chain theory. For more details, we refer to ?. We say that a Markov chain (X_n) with transition kernel P is ϕ -irreducible if there exists a probability measure ϕ such that

$$\phi(A) > 0 \text{ implies that } \Pr(X_n \in A \text{ for some } n | X_0 = x) > 0 \text{ (for all } x \in \mathcal{S}). \quad (2.5)$$

We say that a Markov chain is aperiodic if there does not exist a partition $\mathcal{S} = \mathcal{S}_1 \cup \dots \cup \mathcal{S}_d$ with $d \geq 2$ such that $\Pr(x, \mathcal{S}_{i+1}) = 1$ for all $x \in \mathcal{S}_i$, $i = 1, \dots, d-1$ and $\Pr(x, \mathcal{S}_1) = 1$ for all $x \in \mathcal{S}_d$.

And a Markov chain (X_n) with transition P is ergodic if

$$\|P^n(x, \cdot) - \pi(\cdot)\|_{var} \xrightarrow{n \rightarrow \infty} 0 \quad (2.6)$$

for π -a.e $x \in \mathcal{S}$, where $\|\mu\|_{var} := \frac{1}{2} \sup_{|f| \leq 1} \left| \int \mu(dy) f(y) \right|$ is the total variation norm of a signed measure μ . It is well known that if a ϕ -irreducible Markov chain admits an invariant distribution, and is aperiodic then it is ergodic.

Geometric ergodicity essentially says that the convergence in (2.6) takes place at a geometric rate. More precisely, a Markov chain with transition kernel P is geometrically ergodic if there is $\rho < 1$ and a function $V : \mathcal{X} \rightarrow [1, \infty]$ finite π -almost everywhere such that:

$$\|P^n(x, \cdot) - \pi(\cdot)\|_{var} \leq V(x) \rho^n \quad (2.7)$$

for π -a.e $x \in \mathcal{S}$.

The transition kernel P of the chain (X_n) induces a linear bounded operator K on $L^2(\pi)$ the space of real-valued square integrable functions defined on $\mathcal{S}$ which is given by:

$$Kf(x) := \int f(y) P(x, dy). \quad (2.8)$$

Instead of the operator K defined in (2.8), we shall mainly work with K_0 , the restriction of K to $L_0^2(\pi) = \{f \in L^2(\pi) : \pi(f) := \int f d\pi = 0\}$. It has been shown that (confer ? and ?) for reversible Markov chains, geometric ergodicity is equivalent to:

$$\|K_0 f\| \leq \rho \|f\|, \quad (2.9)$$

for some $\rho < 1$.

Define the spectrum of K_0 by $\sigma(K_0) := \{\lambda \in \mathbb{R} : K_0 - \lambda I \text{ is non invertible}\}$ where I is the identity operator of $L_0^2(\pi)$, and write $r(K_0) := \sup \{|\lambda| : \lambda \in \sigma(K_0)\}$ for the spectral radius of K_0 . Then it is well known that (2.9) is equivalent to $r(K_0) = \|K_0\| < 1$, where $\|K_0\|$ is the norm of the operator K_0 defined by $\|K_0\| := \sup_{\|f\| \leq 1} \|K_0 f\|$. Whenever $\|K_0\| < 1$, we say that the chain has a spectral gap.

In the study of the spectrum of K_0 , we make the distinction between $\sigma_d(K_0)$, the discret spectrum of K_0 and $\sigma_{ess}(K_0) = \sigma(K_0) \setminus \sigma_d(K_0)$ the essential spectrum of K_0 . the discret spectrum $\sigma_d(K_0)$ is

defined as those λ in $\sigma(K_0)$ which are eigenvalues of K_0 and are isolated point of $\sigma(K_0)$ and such that $\dim \ker (K_0 - \lambda) < \infty$. We shall also denote by

$$\text{ess-ran}(r) := \{\lambda \in \mathbb{R} : \pi \{x : |r(x) - \lambda| < \epsilon\} > 0 \text{ for all } \epsilon > 0\}$$

the essential range of r . If $\text{ess-inf}(r)$ (respectively $\text{ess-sup}(r)$) is the essential (with respect to π) infimum (respectively essential supremum) of the function r , it is easily seen that $\text{ess-ran}(r) \subseteq [\text{ess-inf}(r), \text{ess-sup}(r)]$ and that both $\text{ess-inf}(r)$ and $\text{ess-sup}(r)$ belong to $\text{ess-ran}(r)$.

The following result from ? will be useful later.

Proposition 2.1. *Suppose that P is an ergodic transition kernel on $(\mathcal{S}, \mathcal{F}, \pi)$ with invariant distribution π . Then K_0 as defined above has no eigenvalue with absolute value 1.*

In the case of the Metropolis-Hastings algorithm, the operator K_0 acts on $f \in L_0^2(\pi)$ as:

$$K_0 f(x) = \int f(y) P(x, dy) \tag{2.10}$$

$$= M_r f(x) + U f(x). \tag{2.11}$$

With $M_r f(x) = r(x)f(x)$ and $U f(x) = \int \alpha(x, y) \omega(x, y) f(y) \pi(dy)$.

In other words, the Metropolis-Hastings operator is a multiplication operator perturbed by an integral operator. The simplest way to study the spectrum of such operator is to make U a compact operator. This will be done through the following condition:

Assumption 2.1. The operator U is a compact $L_0^2(\pi)$ operator.

The following lemma gives a sufficient condition on the proposal transition kernel Q for this compactness condition to hold.

Lemma 2.1. *If*

$$\int Q(x, dy) Q(y, dx) < \infty, \tag{2.12}$$

then U is an Hilbert-Schmidt operator, thus is compact.

Proof. It is sufficient to check that if (2.12) holds then

$$\int \alpha^2(x, y) \omega^2(x, y) \pi(dx) \pi(dy) < \infty.$$

It follows from $\alpha(x, y) \omega(x, y) = \min(\omega(x, y), \omega(y, x))$, that $\alpha^2(x, y) \omega^2(x, y) \leq \omega(x, y) \omega(y, x)$. Therefore:

$$\alpha^2(x, y) \omega^2(x, y) \pi(dx) \pi(dy) \leq Q(x, dy) Q(y, dx).$$

□

The next theorem is the main result of this section. We have used the Weyl's perturbation theorem (?) together with some basic Hilbert space operator theory to show that K_0 has a spectral gap when $\text{ess-sup}(r) < 1$.

Theorem 2.1. *Suppose that the compacity condition on U holds. Suppose also that the Markov chain generated by the Metropolis-Hastings algorithm (algorithm 2.1) with proposal kernel Q is ergodic. Then it has a spectral gap if and only if $\text{ess-sup}(r) < 1$. The essential supremum being taken with respect to π .*

Proof. Since U is a compact operator, by the Weyl's theorem ($\sigma_{\text{ess}}(K_0) = \sigma_{\text{ess}}(M_r)$). The spectrum of the multiplication operator is well known (see ? example 2.6 page 271). $\sigma(M_r) = \text{ess-ran}(r)$ and λ is an eigenvalue for M_r if and only if $\pi\{y : r(y) = \lambda\} > 0$ and the indicator function of the set $\{y : r(y) = \lambda\}$ is an associated eigenfunction. Thus

$$\begin{aligned}\sigma_{\text{ess}}(M_r) &\subseteq \text{ess-ran}(r), \\ &\subseteq [\text{ess-inf}(r), \text{ess-sup}(r)].\end{aligned}$$

Since K_0 is self-adjoint, either $\|K_0\| \in \sigma(K_0)$ or $-\|K_0\| \in \sigma(K_0)$ (? section 34, theorem 2) and $\sigma(K_0)$ is bounded by $\|K_0\|$. Suppose that $\text{ess-sup}(r) < 1$. Then if $\text{ess-sup}(r) < \|K_0\|$, $\|K_0\| \in \sigma_d(K_0)$ (or $-\|K_0\| \in \sigma_d(K_0)$). But for an ergodic chain, we know from proposition (2.1) that ± 1 cannot be eigenvalues for K_0 . Thus $\|K_0\| < 1$.

The necessary part is proposition 5.1 of ?. □

As, we mentioned above, the necessary part of theorem (2.1) holds even without the compacity condition (2.1) (?).

In practice, it is not very hard to construct an ergodic MH algorithm. The following proposition is adapted from ? and ? and gives simple conditions under which the MH kernel is ergodic. Weaker conditions are possible.

Proposition 2.2. *Suppose that there is a set $A \in \mathcal{S}$ with $\pi(A) > 0$ and $\varepsilon > 0$ such that:*

$$\omega(x, y) > 0 \text{ for all } x, y \in \mathcal{S}, \tag{2.13}$$

and

$$\omega(x, y) > \varepsilon \text{ for all } x, y \in A. \tag{2.14}$$

Then the MH kernel is ergodic.

Condition (2.13) is not a restricted requirement and will be satisfied in many situations. If $\omega(x, y)$ is continuous and satisfy (2.13), then (2.14) also will be satisfied and we can take A to be any non empty compact subset of $\mathcal{S}$.

In the case of the IMH algorithm, (2.12) of lemma (2.1) is always satisfy. Therefore we have the following well known result on the geometric ergodicity of the IMH algorithm (?, ?, ? ?).

Corollary 2.1. *Let r be the probability of rejection of the IMH chain as given by equation (2.2). Then $\text{ess-inf}(r) = 0$ and $\text{ess-sup}(r) = 1 - \text{ess-inf}(\omega)$. Therefore the IMH algorithm has a spectral gap if and only if $\text{ess-inf}(\omega) > 0$.*

Note that by proposition 2.2, $\text{ess-inf}(\omega) > 0$ implies that the IMH algorithm is ergodic. In fact, more can be said about the spectrum of the IMH chain. We shall prove the following:

Theorem 2.2. For the IMH algorithm, $\sigma(K_0) \subseteq \text{ess-ran}(r)$. The equality holds if for all $\alpha \in \text{ess-ran}(r)$, $\pi\{y : r(y) = \alpha\} = 0$ or if $\text{ess-ran}(r)$ has no isolated point.

Proof. We know from theorem (2.1) that $\sigma_{\text{ess}}(K_0) \subseteq \text{ess-ran}(r)$. If for all $\alpha \in \text{ess-ran}(r)$, $\pi\{y : r(y) = \alpha\} = 0$, then K_0 has no eigenvalue, and if $\text{ess-ran}(r)$ has no isolated point, then K_0 has no isolated eigenvalue. In both cases, $\sigma_{\text{ess}}(K_0) = \text{ess-ran}(r)$. It remains to show that K_0 has no eigenvalue outside $\text{ess-ran}(r)$.

First note that for $f \in L_0^2(\pi)$,

$$\begin{aligned} \int \alpha(x, y) \omega(y) f(y) \pi(dy) &= \int_{\{y: \omega(x) \geq \omega(y)\}} \omega(y) f(y) \pi(dy) \\ &\quad + \int_{\{y: \omega(x) < \omega(y)\}} \omega(x) f(y) \pi(dy) \\ &= - \int_{\{y: \omega(y) \leq \omega(x)\}} (\omega(x) - \omega(y)) f(y) \pi(dy). \end{aligned} \quad (2.15)$$

Now, take $\lambda \notin \text{ess-ran}(r)$ and suppose that there is a none zero $f_0 \in L_0^2(\pi)$ such that $K_0 f_0(x) = \lambda f_0(x)$. Using equation (2.15) in equation (2.11), this is equivalent to:

$$\int_{\{y: \omega(y) \leq \omega(x)\}} \frac{\omega(x) - \omega(y)}{r(x) - \lambda} f_0(y) \pi(dy) = f_0(x). \quad (2.16)$$

Consider T the operator $Tf(x) = \int_{\{y: \omega(y) \leq \omega(x)\}} \frac{\omega(x) - \omega(y)}{r(x) - \lambda} f(y) \pi(dy)$. Then equation (2.16) says that 1 is an eigenvalue for T with eigenfunction f_0 . We shall now show that this is not possible and the theorem will be proved.

Note $\underline{\omega} = \text{ess-inf}(\omega(x))$ and $\kappa = \text{ess-inf}(|r(x) - \lambda|) > 0$. Since $f_0 \neq 0$, we can find $u > \underline{\omega}$ sufficiently large such that $f_0 \neq 0$ on $\{x \in \mathcal{S} : \underline{\omega} \leq \omega(x) < u\}$. For any partition $I_n = (u_0 \leq u_1 \leq \dots \leq u_n)$ of $[\underline{\omega}, u]$, with $u_0 = \underline{\omega}$ and $u_n = u$, we note $D(u_i) = \{x \in \mathcal{S} : u_{i-1} \leq \omega(x) < u_i\}$ and $L_i^2(\pi) = \{h \in L_0^2(\pi) : h(x) = 0 \text{ for } x \notin D(u_i)\}$, $i = 1, \dots, n$. $L_i^2(\pi)$ is an Hilbert space as a closed subspace of $L_0^2(\pi)$. Let M_{D_i} be the multiplication operator by $\chi_{D(u_i)}$, the indicator function of the set $D(u_i)$. For $h \in L_0^2(\pi)$, we write h_{D_i} for $h\chi_{D(u_i)} = M_{D_i}h$. With these notations, it is easy to see that equation (2.16) implies:

$$\begin{cases} M_{D_1} T M_{D_1} f_{0,D_1} &= f_{0,D_1} \\ M_{D_2} T M_{D_2} f_{0,D_2} &= f_{0,D_2} - M_{D_2} h_2 \\ \vdots & \\ M_{D_n} T M_{D_n} f_{0,D_n} &= f_{0,D_n} - M_{D_n} h_n \end{cases} \quad (2.17)$$

where $h_j(x) = \sum_{k=1}^{j-1} \int_{D(u_k)} \frac{\omega(x) - \omega(y)}{r(x) - \lambda} f_0(y) \pi(dy)$, for $j = 2, \dots, n$. Equation (2.17) implies that $r(M_{D_i} T M_{D_i}) \geq 1$ for at least one i in $1, \dots, n$.

But for $g \in L_i^2(\pi)$ with $\|g\| = 1$, and using the Cauchy-Schwartz inequality, we can write:

$$\begin{aligned} \|M_{D_i} T M_{D_i} g\|^2 &= \int_{D(u_i)} \left\{ \int_{\{y: \omega(y) \leq \omega(x)\}} \frac{\omega(x) - \omega(y)}{r(x) - \lambda} g_{D_i}(y) \pi(dy) \right\}^2 \pi(dx) \\ &\leq \left(\frac{u_i - u_{i-1}}{\kappa} \right)^2 \int_{D(u_i)} g^2(y) \pi(dy) \\ &\leq \left(\frac{u_i - u_{i-1}}{\kappa} \right)^2 \end{aligned}$$

Therefore $\|M_{D_i} T M_{D_i}\| \leq (u_i - u_{i-1}) / \kappa$. From this, by taking a partition $I_n = (u_0 \leq u_1 \leq \dots \leq u_n)$ with $\max_{1 \leq i \leq n} (u_i - u_{i-1}) < \kappa$, we can write for $i = 1, \dots, n$:

$$\begin{aligned} r(M_{D_i} T M_{D_i}) &= \lim_{n \rightarrow \infty} \|(M_{D_i} T M_{D_i})^n\|^{\frac{1}{n}} \\ &\leq \|M_{D_i} T M_{D_i}\| \\ &\leq \frac{u_i - u_{i-1}}{\kappa} \\ &< 1. \end{aligned}$$

Thus the theorem is proved. $\square$

Remark 2.1. 1. A typical case for theorem (2.2) will be that $\mathcal{S} = \mathbb{R}^d$ and π and Q are absolutely continuous with respect to the Lebesgue measure: $\pi(dx) = \pi(x)dx$ and $Q(dx) = q(x)dx$. If in addition, $\pi(x)$ and $Q(x)$ are positive and continuous, then $r(x)$ is also continuous and $\sigma(K_0) = [0, 1 - \text{ess-inf}(\omega)]$.

2. One well known consequence of theorem (2.2) is that it is not possible to take advantage of the correlation of the Markov chain to decrease the asymptotic variance of the Monte Carlo estimate. Suppose that (X_n) is a geometrically ergodic Markov chain with stationary distribution π , generated using the IMH algorithm. For any $f \in L_0^2(\pi)$, writting e_f to denote the spectral measure of K_0 on f and $\rho := 1 - \text{ess-inf}(\omega)$, we have:

$$\lim_{n \rightarrow \infty} \frac{1}{n} E \left(\sum_{k=0}^{n-1} f(X_k) \right)^2 = \int_0^\rho \frac{1+\lambda}{1-\lambda} e_f(d\lambda) \geq \|f\|^2.$$

3. The Restricted Metropolis Algorithm.

In practice, the RWMH algorithm remains one of the most used MH algorithm. But for many common distribution, the RWMH algorithm will fail to convergence at a geometric rate. In fact, as we mentioned in the introduction, exponential or lighter tail is necessary for the RWMH algorithm to be geometrically ergodic (? , ?). Here, we propose a MH algorithm that behaves as the RWMH on a compact set. We show in theorem 3.1 that the proposed algorithm is geometrically ergodic whenever the target density's tail decays at least as fast as the proposal density's tail. In this section, we take $\mathcal{S} = \mathbb{R}^d$ and assume that π is absolutely continuous with respect to the Lebesgue measure and we write: $\pi(dx) = \pi(x)dx$. Moreover, we also assume that the function π is positive and bounded on compact sets. Let $Q(x, dy) = q(|y - x|)dy$ be a proposal kernel. We assume that q is positive, continuous and $q(r)$ is nonincreasing for $r > 0$ sufficiently large. Let Δ be a compact, convex and non empty subset of $\mathcal{S}$. We define $p : \mathcal{S} \rightarrow \mathcal{S}$ by $p(x) = \text{argmin}_{y \in \Delta} \{|y - x|\}$ the orthogonal projection on Δ . We propose the following Metropolis-Hastings move that we call restricted RWMH:

Algorithm 3.1. 1. Start the chain at any point $x \in \mathcal{S}$: $X_0 = x$.

2. Conditionally to X_i , generate $Y \sim Q(p(X_i), \cdot)$ and generate $U \sim \mathcal{U}(0, 1)$.

3. If $U \leq \alpha(X_i, Y)$ then set $X_{i+1} = Y$, otherwise set $X_{i+1} = X_i$. Where $\alpha(x, y) = \min \left(1, \frac{\pi(y)q(|x-p(y)|)}{\pi(x)q(|y-p(x)|)} \right)$.

On Δ , this algorithm behaves like the RWMH algorithm with kernel $Q(x, \cdot)$. For $x \notin \Delta$, the proposal is taken from $Q(p(x), \cdot)$ where $p(x)$ is the projection of x on Δ . Typically, if x_0 is any

position parameter of π , one can take $\Delta = B(x_0, R)$, the closed ball centered at x_0 with radius $R > 0$ taking sufficiently large for Δ to include all the essential features of π . In this case, $p(x) = x_0 + (\min(1, R/|x - x_0|))(x - x_0)$ is easy to compute.

Clearly, the transition density becomes:

$$P(x, y)dy = \alpha(x, y)q(|y - p(x)|)dy + r(x)\delta_x(dy),$$

where

$$\alpha(x, y) = \min\left(1, \frac{\pi(y)q(|x - p(y)|)}{\pi(x)q(|y - p(x)|)}\right),$$

and

$$r(x) = 1 - \int \alpha(x, y)q(|y - p(x)|)dy. \quad (3.1)$$

Theorem 3.1. *Suppose that:*

$$\liminf_{|x| \rightarrow \infty} \frac{q(|x|)}{\pi(x)} > 0. \quad (3.2)$$

Then the Restricted RWMH algorithm (3.1) is geometrically ergodic.

Proof. From the conditions satisfied by π and q , it follows that for any compact set $A \in \mathcal{F}$, there is $\varepsilon > 0$ such that $\omega(x, y) = \frac{q(|y - p(x)|)}{\pi(y)} > \varepsilon$ for all $x, y \in A$. Therefore from proposition (2.2) the restricted RWMH is ergodic.

To finish the proof, it remains to show that the compacity condition in Assumption (2.1) is satisfied and next prove that r , the rejection probability as given by (3.10) is bounded away from 1.

Showing the compacity condition 2.1 amounts to show that

$$I := \int dx \int q(|y - p(x)|)q(|x - p(y)|)dy < \infty.$$

Take $M > 0$ such that q is nonincreasing on $\{r > M\}$, and write $\tau := \sup\{|z|, z \in \Delta\} < \infty$. Then for $|y| > M + \tau$ and for $x \in \mathcal{S}$, $|y - p(x)| \geq |y - p(y)| \geq |y| - \tau > M$. This implies that for $x \in \mathcal{S}$, and for $|y| > \tau + M$, $q(|y - p(x)|) \leq q(|y - \tau|)$.

By writing $A := \int_{\{|x| \leq \tau + M\}} dx \int_{\{|y| \leq \tau + M\}} q(|y - p(x)|)q(|x - p(y)|)dy < \infty$, we therefore have:

$$\begin{aligned} I &= \int_{\{|x| \leq \tau + M\}} dx \int_{\{|y| \leq \tau + M\}} q(|y - p(x)|)q(|x - p(y)|)dy \\ &\quad + \int_{\{|x| \leq \tau + M\}} dx \int_{\{|y| > \tau + M\}} q(|y - p(x)|)q(|x - p(y)|)dy \\ &\quad + \int_{\{|x| > \tau + M\}} dx \int_{\{|y| \leq \tau + M\}} q(|y - p(x)|)q(|x - p(y)|)dy \\ &\quad + \int_{\{|x| > \tau + M\}} dx \int_{\{|y| > \tau + M\}} q(|y - p(x)|)q(|x - p(y)|)dy \\ &\leq A + \int_{\{|x| > \tau + M\}} q(|x - \tau|)dx \left(\int_{\{|x| \leq \tau + M\}} q(|x - p(y)|)dx + \int_{\{|y| \leq \tau + M\}} q(|y - p(x)|)dy \right) \\ &\quad + \left(\int_{\{|y| > \tau + M\}} q(|x - \tau|)dx \right)^2 \\ &< \infty. \end{aligned}$$

Using condition (3.2) and the same arguments as above, we can see that:

$$\inf_{z \in \Delta, x \in \mathcal{S}} \frac{q(|x - z|)}{\pi(x)} = \eta > 0,$$

and

$$\sup_{x \in \mathcal{S}} \frac{q(|x - p(x)|)}{\pi(x)} = \eta_0 < \infty.$$

Therefore we have:

$$\begin{aligned} \alpha(x, y) &= \min \left(1, \frac{\pi(y)q(|x - p(y)|)}{\pi(x)q(|y - p(x)|)} \right) \\ &\geq \min \left(1, \frac{q(|x - p(y)|)/\pi(x)}{q(|y - p(y)|)/\pi(y)} \right) \\ &\geq \frac{\eta}{\eta_0}, \end{aligned}$$

for $|y| > \tau + M$. It follows that:

$$\begin{aligned} \int \alpha(x, y)q(|y - p(x)|)dy &\geq \frac{\eta}{\eta_0} \int_{\{|y| > \tau + M\}} q(|y - p(x)|)dy \\ &\geq \frac{\eta^2}{\eta_0} \pi(\{|y| > \tau + M\}). \end{aligned}$$

This implies that $r(x) \leq 1 - \frac{\eta^2}{\eta_0} \pi(\{|y| > \tau + M\}) < 1$. And the theorem is proved. $\square$

Remark 3.1. With condition (3.2) of theorem (3.1), the IMH algorithm with proposal distribution q is also geometrically ergodic. But depending on the features of the density π , the mixing rate of this IMH chain may be slow. The restricted version of the RWMH algorithm retains the geometric ergodicity of the IMH but will still explore Δ typically like the RWMH algorithm.

Actually, the convergence rate of the Restricted RWMH is much more similar to that of the IMH than what theorem (3.1) can reveal. In the next theorem, using Markov chain techniques, we show that the restricted version of RWMH algorithm is uniformly geometrically ergodic. We also show, in the same way as ? did for the IMH, that if $\liminf_{|x| \rightarrow \infty} \frac{q(|x|)}{\pi(x)} = 0$, the RWMH algorithm can still have a polynomial rate of convergence.

For $x_0 \in \Delta$, write as above $\omega(x_0, x) = \frac{q(|x - x_0|)}{\pi(x)}$ and note:

$$\delta(x_0) := \inf_{y \in \Delta} \omega(x_0, y), \tag{3.3}$$

$$\delta := \sup_{x_0 \in \Delta} \delta(x_0).$$

Also write:

$$\theta(x_0) := \inf_{x, y \in \mathbb{R}^d} \frac{q(|y - p(x)|)}{q(|y - x_0|)} > 0, \tag{3.4}$$

$$\tau(x_0) := \sup_{x, y \in \mathbb{R}^d} \frac{q(|y - p(x)|)}{q(|y - x_0|)} < \infty, \tag{3.5}$$

and

$$\theta := \sup_{x_0 \in \Delta} \theta(x_0) > 0.$$

Note that if $\delta > 0$ then $\delta(x_0) > 0$ for all (not just for some) $x_0 \in \Delta$. And of course if for some $x_0 \in \Delta$, $\delta(x_0) > 0$ then $\delta > 0$.

Theorem 3.2. (i) *If $\delta > 0$, then*

$$\|P^n(x, \cdot) - \pi(\cdot)\|_{TV} \leq (1 - \theta\delta)^n, \quad (3.6)$$

for every $x \in \mathbb{R}^d$.

(ii) *Assume that $\delta = 0$. Then P cannot converge at a geometric rate.*

(iii) *Assume that $\delta(x_0) = 0$ for some $x_0 \in \Delta$. If for some $r > 0$,*

$$\pi(A_\varepsilon) = O(\varepsilon^{1/r}) \text{ for } \varepsilon \rightarrow 0, \quad (3.7)$$

where $A_\varepsilon = \{x \in \mathbb{R}^d : \omega(x_0, x) \leq \varepsilon\}$, then for $r < s < r + 1$, we have

$$r_s(n) \|P^n(x, \cdot) - \pi(\cdot)\|_{TV} \longrightarrow 0 \text{ as } n \rightarrow \infty, \quad (3.8)$$

with $r_s(n) = (n + 1)^{\frac{s}{r}-1}$.

Proof. (i) Because $\delta > 0$, we can find and fix $x_0 \in \Delta$ such that $\delta(x_0) > 0$. The transition kernel of the RM algorithm is:

$$P(x, dy) = \alpha(x, y)q(|y - p(x)|)dy + r(x)\mathbb{1}_{\{x\}}(dy), \quad (3.9)$$

where

$$\alpha(x, y) = \min \left(1, \frac{\pi(y)q(|x - p(y)|)}{\pi(x)q(|y - p(x)|)} \right),$$

and

$$r(x) = 1 - \int \alpha(x, y)q(|y - p(x)|)dy. \quad (3.10)$$

If $\alpha(x, y) = 1$, $P(x, dy) \geq \frac{q(|y - p(x)|)}{\pi(y)}\pi(dy) \geq \theta(x_0)\delta(x_0)\pi(dy)$. In the same way, if $\alpha(x, y) = \frac{\pi(y)q(|x - p(y)|)}{\pi(x)q(|y - p(x)|)}$, then $P(x, dy) \geq \frac{q(|x - p(y)|)}{\pi(x)}\pi(dy) \geq \theta(x_0)\delta(x_0)\pi(dy)$.

In any case, we have $P(x, dy) \geq \theta(x_0)\delta(x_0)\pi(dy)$ for $x \in \mathbb{R}^d$, meaning that the whole space is a small set for P . Therefore following from ? theorem 16.1., we have: $\|P^n(x, \cdot) - \pi(\cdot)\|_{TV} \leq (1 - \theta(x_0)\delta(x_0))^n$, for all $x \in \mathbb{R}^d$ which implies that:

$$\begin{aligned} \|P^n(x, \cdot) - \pi(\cdot)\|_{TV} &\leq \inf_{x_0 \in \Delta} (1 - \theta(x_0)\delta(x_0))^n \\ &\leq (1 - \theta\delta)^n. \end{aligned}$$

(ii) Since $\delta = 0$, $\delta(x_0) = 0$ for all $x_0 \in \Delta$. We shall show that if for some $x_0 \in \Delta$, we have $\delta(x_0) = 0$, then $\text{ess-sup } r(x)$, the probability of rejection of the chain is 1. Then proposition 5.1 of ? helps us to conclude that P cannot be geometrically ergodic. We do this very much like ?, theorem 2.1.

We fix $x_0 \in \Delta$ such that $\delta(x_0) = 0$ and we write $D_n = \{x \in \mathbb{R}^d : \omega(x_0, x) \leq \frac{1}{n}\}$. Since $\delta(x_0) = 0$, $\pi(D_n) > 0$ for all $n \geq 1$. For $x \in \mathbb{R}^d$, we note $A_x = \left\{y \in \mathbb{R}^d : \frac{\pi(y)q(|x-p(y)|)}{\pi(x)q(|y-p(x)|)} \geq 1\right\}$ and $R_x = \left\{y \in \mathbb{R}^d : \frac{\pi(y)q(|x-p(y)|)}{\pi(x)q(|y-p(x)|)} < 1\right\}$. We have:

$$\begin{aligned} r(x) &= \int_{R_x} (1 - \alpha(x, y))q(|y - p(x)|)dy \\ &= 1 - \int_{A_x} q(|y - p(x)|)dy - \int_{R_x} \alpha(x, y)q(|y - p(x)|)dy. \end{aligned} \quad (3.11)$$

Take $y \in A_x$, we have: $1 \leq \frac{\pi(y)q(|x-p(y)|)}{\pi(x)q(|y-p(x)|)} \leq \frac{\omega(x_0, x)}{\omega(x_0, y)} \frac{\theta(x_0)}{\tau(x_0)}$. Therefore, for $x \in D_n$ and $y \in A_x$, $\omega(x_0, y) \leq \frac{\theta(x_0)}{\tau(x_0)} \frac{1}{n}$. This implies that

$$\int_{A_x} q(|y - p(x)|)dy \leq \int_{A_x} \tau(x_0)\omega(x_0, y)\pi(dy) \leq \theta(x_0)\frac{1}{n}, \quad (3.12)$$

for all $x \in D_n$. In the same way, for $x \in D_n$ and $y \in R_x$, we have:

$$\begin{aligned} \int_{R_x} \alpha(x, y)q(|y - p(x)|)dy &= \int_{R_x} \frac{\pi(y)q(|x-p(y)|)}{\pi(x)}dy \\ &\leq \omega(x_0, x)\tau(x_0) \\ &\leq \tau(x_0)\frac{1}{n}. \end{aligned} \quad (3.13)$$

Equations (3.12) and (3.13) in (3.11) gives:

$$r(x) \geq 1 - (\theta(x_0) + \tau(x_0))\frac{1}{n},$$

for all $x \in D_n$, $n \geq 1$. Since $\pi(D_n) > 0$ for all $n \geq 1$, it follows that $\text{ess-sup } r(x)$ (with respect to π) is 1.

(iii) First, note that $P(x, dy) \leq P_0(x, dy)$ where

$$P_0(x, dy) = \alpha_0(x, y)\tau\omega(x_0, y)\pi(dy) + \left[1 - \theta(x_0) \int \alpha_0(x, y)\omega(x_0, y)\pi(dy)\right] \mathbb{1}_{\{x\}}(dy),$$

and $\alpha_0(x, y) = \min\left(1, \frac{\tau(x_0)}{\theta(x_0)} \frac{\omega(x_0, x)}{\omega(x_0, y)}\right)$.

As in theorem 5.3 of ?, it can be shown that there exist constant $0 < c, b < \infty$ such that P_0 satisfies the drift condition $P_0 V \leq V - V^\alpha + b\mathbb{1}_C$ for some petite set C (for P), $\alpha = 1 - \frac{r}{s}$ and $V(x) = \omega(x_0, x)^{-s/r}$. Therefore the same drift condition applies to P and the result follows from theorem 3.6 of ?.

□

Example 3.1. In this simulation study, we illustrate the benefits of restricting the RWMH in terms of the existence of the Central Limit Theorem. We say that the Central Limit Theorem holds for a function h if there exists $0 < \sigma_h^2 < \infty$ such that:

$$\sqrt{n}(S_n(h) - \pi(h)) \xrightarrow{w} N(0, \sigma_h^2),$$

where $S_n(h) = \frac{1}{n} \sum_{i=1}^n h(X_i)$. It is well known that for a reversible geometrically ergodic Markov chain, the Central Limit Theorem holds for any $h \in L^2(\pi)$, see ? for example.

Note $t(\nu)$ the student distribution on $\mathbb{R}$ with ν degree of freedom. Clearly, from theorem (2.2) of ?, the RWMH with target density $t(\nu)$ is not geometrically ergodic. But using a proposal distributed as $t(r)$ with $r \leq \nu$, we know from theorem (3.1) that the restricted version of the RWMH presented above is geometrically. Thus the Central Limit Theorem holds for any h with $\pi(h^2) < \infty$.

We use $h(x) = |x|$, $r = 0.5$ and $\nu = 3, 4$. For each of these 4 combinations ($\nu = 3, 4$ times ordinary and restricted RWMH), we ran $N = 1,000$ independent chains starting at 0 each with length $n = 1,000,000$. For the restricted versions, we have used $\Delta = B(0, R)$ where R is chosen such that $\Pr(|t(\nu)| < R) = 0.9$. For each of these 4 combinations, the i -th run is used to estimate $\pi(|x|)$ by $S^i(|X|) = \frac{1}{n} \sum_{k=1}^n |X_k^{(i)}|$. Figures 1 and 2 show the QQplot versus the normal distribution and the histogram of the normalised empirical sum $S^i(|X|) - \frac{1}{N} \sum_{i=1}^N S^i(|X|)$.

For $\nu = 3$, as expected, the Central Limit Theorem seems to hold for $h(x) = |x|$ in the case of the restricted version of the RWMH algorithm, while it clearly fails for the ordinary RWMH. For $\nu = 4$, the Central Limit Theorem seems to hold for both versions. This agrees with theorem 3.1 and theoretical results obtained by ? on the RWMH algorithm.

Figure 1: QQ-plot and histogram of 1,000 normalised ergodic averages of the function $h(x) = |x|$. Target density $t(3)$, proposal density $t(0.5)$. The KS value is the p-value of the Kolmogorov-Smirnov test of normality.

Figure 2: QQ-plot and histogram of 1,000 normalised ergodic averages of the function $h(x) = |x|$. Target density $t(4)$, proposal density $t(0.5)$. The KS value is the p-value of the Kolmogorov-Smirnov test of normality.

4. Remarks

From a practical point of view, there are many problems remaining to be solved with this restricted version of the RWMH algorithm. In our sense, the most important is how to choose the set Δ . In many situations there may not be enough information available on π to make a good choice for Δ . One natural possibility is to use some adaptive schemes to update Δ during the simulation. See for example ?, ?. The regenerative framework for adaption proposed by ? seems particularly interesting in this respect as we suspect the restricted RWMH algorithm to have more frequent regenerations than the RWMH algorithm. But more work need to be done in this direction.